

Evaluation of Algeria's energy potential: hybrid energy system analysis and sizing with HOMER pro.

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Abstract— Renewable energies play an important role in slowing climate change, decreasing our reliance on fossil fuels, and guaranteeing a sustainable future for the next generations. In this article, Homer Pro was used to obtain the best size for a hybrid energy system, with the aim of reducing the cost of this system while increasing its reliability to meet the load demand in a continuous way. This system combined multiple energy sources include wind turbines, photovoltaic panels and battery backup. This combination enhances the efficiency and reliability of power generation, allowing for the optimal utilization of renewable energies. Six different regions of Algeria were chosen to size this system. Each region differs from the other in meteorological characteristics, allowing us to study the potential of energy from renewable sources in the country for the use of hybrid energy systems. The obtained result show that reliance on one energy source raises the system's cost. As consequently, the hybrid system helps to improve reliability of the system while also lowering costs. Furthermore, meteorological data indicates that Algeria is suitable for the utilization of renewable energies due to the diversity and abundance of its energy sources.

Keywords—Hybrid energy system, Size, HOMER pro, Photovoltaic, Wind, Battery.

I. INTRODUCTION

Global energy demand is increasing at an accelerated rate during this period of rapid urbanization and industrialization [1]. However, conventional energy sources (mainly fossil fuels) have serious adverse effects on the environment in addition to their rising costs. The combustion of fossil fuels releases carbon emissions that contribute significantly to climate change and global warming, hence exacerbating natural catastrophes, modifying ecosystems, and endangering biodiversity [2], [3]. Furthermore, the production and transportation of fossil fuels cause ecological harm, such as oil spills [4]. In light of these conditions, renewable energy sources (RES) become not only the better option but also essential one to protect the environment. They have two major benefits: since they are never-ending, they ensure long-term sustainability, and because they are naturally cleaner, they significantly reduce carbon emissions [3]. On the other hand, Algeria has a large area of 2.38 million square kilometers, which makes it a diverse and abundant region of renewable energy in North Africa, from the Sahara Desert, which is rich in solar energy, to the coasts of the Mediterranean Sea, which is rich in wind energy [5], [6].

Hybrid energy systems (HES) represent an innovative solution in the field of renewable energy. By definition, a hybrid energy system combines more than one source of energy in order to generate electricity. In the context of a solar-wind-battery HES, it exploits solar energy using photovoltaic cells and wind energy using turbines [7]. The addition of batteries ensures that excess energy produced can be stored for later use, and they are also a solution to the variable nature of renewable energy caused by unpredictable sunlight and wind conditions. Therefore, these systems optimize energy generation and provide a consistent output, making them crucial for areas with unstable weather or for applications demanding constant energy [8].

However, Accurate sizing is essential to maximizing the efficiency of a HES. Determining the best size for each component, from solar panels to wind turbines and batteries, is essential to ensure that every element performs at its peak capacity without the risk of failing to satisfy the load demand [3]. The process involves identifying the best size of HES components to reduce costs and effectively fulfil the demand of the local load. This work could be done by using an optimization method [9], [10]. Moreover, when a hybrid system is appropriately dimensioned, it offers a consistent and reliable energy supply. This characteristic enhances its reliability, particularly in instances when renewable energy sources exhibit unpredictability, so providing a stable solution to the challenges associated with fluctuating RESs and energy demands [9].

Numerous studies can be found on the design and optimization of HESs. Most of these studies use traditional methods, artificial intelligence method, hybrid method or software tools [11]. Kaabeche et al. [12] applied the iterative technique to determine the best sizing of the elements of the HES (PV-wind-battery) designed to provide power to a residential house located in the center of the development of renewable energies located in Bouzaréah, Algeria. The ideal size model's flowchart is illustrated for a simulation of 8760 hours. The authors considered evaluation criteria such as LPSP and LCOE to minimize them in order to choose the best optimal configurations of the HES. In [7], a sizing methodology using the Cuckoo Search algorithm is presented for HESs connected to the grid. Optimal sizing is a multi-objective issue with technical, environmental and economic

criteria. The developed approach is evaluated for both residential and agricultural loads in the province of Ghardaïa, Algeria. The developed approach provides faster convergence, improved accuracy and reduced computation time in comparison to the PSO optimization approach. Medghalchi et al. [3] introduced a new method to study the combination of solar PV and wind HESs with batteries and electrolyzer-fuel cell for energy storage system. The aim is to reduce the cost of the system and increase its reliability. To meet this aim, a novel hybrid optimization approach combining particle swarm optimization and genetic algorithms is presented to optimize the size of the system component. The proposed approach provides more applicable results. Zhang et al. [13] size and optimize a hybrid system using the HOMER PRO software based on the system of a diesel generator, a wind turbine, PV and a battery. From the simulation results, it was proven that the industrial city has sufficient potential to generate energy from wind and solar sources. Also, it was proven by the obtained results that the utilization of photovoltaics and wind is necessary for the system, since the presence of each of them reduces emissions and also maintenance and operating costs.

In this article, a techno-economic study has been introduced and presented to size a HES combined of PV/wind/battery in various provinces in Algeria considering seasonal changes in load demand, alongside solar and wind energy data, as well as component costs and maintenance expenses.

II. METHODOLOGY

The methodology flowchart follows the HOMER pro flowchart, shown in Fig. 1. The process begins with estimating

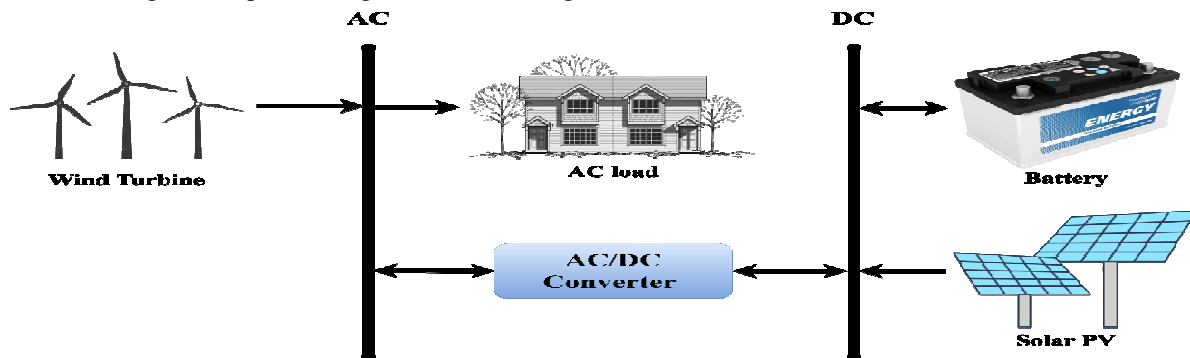


Fig. 1. Schematic of the proposed HES.

B. HOMER pro presentation

HOMER Pro, standing for Hybrid Optimization of Multiple Energy Resources, is a software tool developed originally by the U.S. National Renewable Energy Laboratory (NREL) with the aim of optimizing microgrid designs, both in scenarios connected to the grid and those that are off-grid [14], [15]. This software is among the important tools in the field of hybrid energy system sizing, modeling, and design. HOMER Pro stands out from other tools due to its ability to model the combination of various energy sources, from classic generators to modern options like wind turbines, solar panels, and battery storage [16]. A

notable feature of HOMER Pro is its sensitivity analysis. This allows users to explore different variables, from changing fuel costs and equipment expenses to renewable energy data, leading to an optimal sizing of the systems [17]. Coupled with this, HOMER Pro can perform multi-year system operation simulations, taking into account challenges such as equipment maintenance and replacement [14]. In addition, HOMER Pro includes an extensive library of numerous pre-defined energy components, which simplifies the process of modeling diverse energy system configurations. HOMER Pro also focuses on the economic indicators of energy systems. It provides economic

A. Proposed system

In this study, the proposed system is a mix of two primary renewable energy sources: wind (wind turbine) and solar (solar PV panels) as shown in the Fig. 1. The purpose of mixing these two sources is to ensure a more stable and consistent power supply, as the availability of sun and wind can vary throughout the day and seasons. The wind turbine is shown on the left side as connected to the AC (Alternating Current) line. Wind turbines typically generate AC electricity. The solar PV panels are connected to the DC (Direct Current) line on the right side. Solar panels produce DC electricity, which often requires conversion to AC for most household and commercial uses. The system also integrates a battery storage unit on the DC side. Batteries store excess electricity generated during peak sunlight or wind hours and release it during periods of low generation, ensuring a steady power supply. Between the AC and DC lines, there is an AC/DC converter. This essential component converts electricity from DC to AC or AC to DC, facilitating the integration of power between the two energy sources, the storage unit and the load.

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evaluations, including metrics such as the total net present cost (NPC) and the levelized cost of energy (LCOE) [18].

The working method of HOMER Pro software is illustrated in Fig. 2. Once details such as microgrid constraints, microgrid control, resource data, economic details and load profile are entered, the software runs the simulations to optimize the hybrid energy system. After that, the results are obtained as outputs, including optimal net present cost, component sizing, energy cost, unserved load, excess electricity, return on investment, and other indicators optimization output data [19].

C. Optimization method

The process of designing an efficient hybrid power system necessitates the application of a suitable optimization method. This ensures that the system is not only efficient but also cost-effective and reliable. Therefore, the optimization approach defines the objective functions and constraints to get the optimal size of the HES. HOMER pro utilizes a derivative-free algorithm to identify the most cost-effective system. The optimization approach uses a modified version of the grid-search technique [14].

1) Objective function

One of the foundational steps in optimization is the definition of objective functions. In the context of energy systems, this function is the reduction of the Levelized Cost of Energy. The LCOE represents the per-unit cost of building and operating a power generating system over its lifetime [18]. It is often expressed in currency per unit of electricity, such as euro per kilowatt-hour (€/kWh). The LCOE provides a means to compare the costs of different configurations of electricity generation on a consistent basis. The formula for LCOE that HOMER Pro uses is represented as [15]:

$$LCOE = \frac{C_{ann,tot} - C_{boiler} H_{served}}{E_{served}} \quad (1)$$

Where:

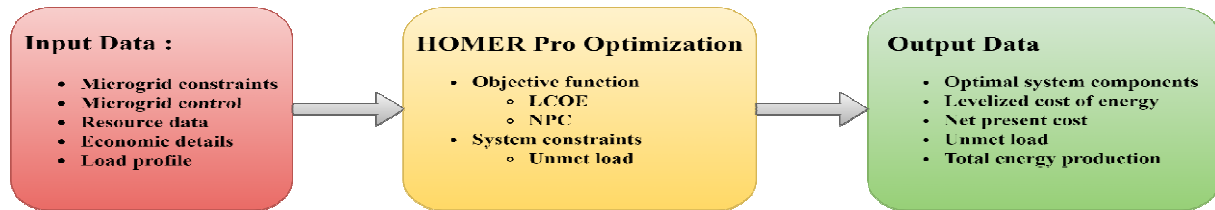


Fig. 2. HOMER Pro Flowchart: From Input Data to Optimization and Output Results.

III. RESULT AND DISCUSSION

To determine the best system configuration, the HOMER Pro software was utilized to execute a simulation based on the COE and unmet load values mentioned earlier in this article. The system used in this study is simulated by taking into account the following sensitivity parameters: the value of COE should be less than 1.5, while the value of unmet load should be less than 0.1%.

- $C_{ann,tot}$: Annual total cost for the system.
- C_{boiler} : Boiler marginal cost.
- H_{served} : Total thermal energy demand met over a year.
- E_{served} : Total electrical consumption met.

2) Constraints

Several constraints are defined to ensure the system's feasibility and efficiency. These include: Load demand should always be met, maximum and minimum generation capacities and storage system's state-of-charge limitations [18].

a) Load Demand Should Always be Met

The primary aim of any power system is to ensure its ability to consistently meet the energy needs of its consumers. This constraint is a guarantees that the generated power will always equal or exceed the consumer demand [18].

b) Storage System's State-of-Charge Limitations

Batteries come with upper (fully charged) and lower (fully discharged) state-of-charge boundaries. Exceeding or dropping below these limits can adversely affect the storage's lifetime. By establishing these constraints, the system can ensure the storage component's efficiency, safety, and prolonged lifetime, preventing potential damages or rapid degradation [20].

c) Maximum and Minimum Generation Capacities

Every energy-generating system has its capacity limits. The maximum generation capacity represents the highest energy output the system can produce under peak conditions, while the minimum generation capacity indicates the least energy the system must produce to remain operationally viable. Defining these boundaries is important to protect against potential overloads or underperformances, which in turn ensures the long lifetime of the equipment and optimizes system performance [18].

The data presented in the Table I represents the monthly climate details for six selected Algerian cities: Adrar, Ouargla, Béchar, Sétif, Alger, and Tiaret. This information was collected from the NASA POWER platform, which offers a comprehensive dataset of solar and meteorological data from around the world. This data is structured around three main climatic parameters: Daily Temperature (DT), Daily Radiation (DR), and Wind Speed (WS). Starting with temperature data, a noticeable seasonal shift can be seen. During the summer, temperatures rise,

with Adrar recording a peak of 36.2°C in July, showcasing the harsh desert conditions, and This slightly affects solar power generation. In contrast, Tiaret has 8.3°C in January, presenting the milder winter temperatures. On transitioning to radiation levels, which signify solar intensity and the potential for solar-powered ventures, Ouargla stands out prominently with a remarkable 7.9 kWh/m² in July. This level suggests the city's solar energy prospects and hints at its suitability for agricultural projects that rely on solar energy. In December, Sétif and Algiers have a low 2.1 kWh/m² due to their location and the season. This suggests that other energy sources might be needed during this time. The wind data in these cities is not as straightforward as the temperature and solar numbers. Nevertheless, we can see that Adrar had the highest wind speed among the six cities of 7.9 m/s in July. This means Adrar could use both solar and wind energy. Other cities sometimes have lower wind speeds of 3.6 m/s, showing quieter wind conditions. Finally, we can conclude that Algeria's various regions have diverse climatic conditions, each offering distinct energy potentials.

Cities like Adrar have significant solar and wind energy potential, while other areas may favor one energy source. The data shows that Algeria has many chances to use renewable energy, and we should size and design this energy system based on each area's weather. The Fig. 3 illustrates the daily consumption profile, with the highest demand observed between 17:00 and 21:00 hours. This peak influences the sizing of the system. The profile used in this study indicates a daily average energy consumption of 11.26 kWh.

Table II presents the specifications and costs of components used in this study: PV panel, wind turbine, lead-acid battery, and converter. The table provides essential data for the practical design of a HES. Utilizing the detailed specifications and costs of these components, HOMER pro can accurately size and optimize a hybrid energy system. This information aids in achieving an optimal solution to ensure the system's reliability, cost-effectiveness, and long-term sustainability.

TABLE I. MONTHLY CLIMATE DATA FOR SELECTED ALGERIAN CITIES: DAILY TEMPERATURE (C): DT. DAILY RADIATION (KWH/M²): DR. WIND SPEED (M/S): WS.

Month	Adrar			Ouargla			Béchar			Sétif			Algiers			Tiaret		
	DT	DR	WS	DT	DR	WS	DT	DR	WS	DT	DR	WS	DT	DR	WS	DT	DR	WS
January	12,5	3,7	5,7	11,9	3,0	4,9	7,2	3,5	4,7	8,8	2,3	5,2	10,5	2,4	5,0	8,3	2,5	4,6
February	15,4	4,8	5,5	14,0	4,3	5,1	9,9	4,4	4,8	9,7	3,3	5,3	11,1	3,3	5,1	9,8	3,4	4,9
March	19,6	6,1	6,0	17,7	5,6	5,3	14,1	5,5	5,2	12,0	4,4	5,3	12,9	4,5	5,1	12,7	4,6	4,9
April	24,3	7,1	5,8	22,1	7,0	5,7	18,1	6,7	5,3	14,6	5,4	5,3	15,1	5,6	5,1	15,4	5,5	5,1
May	28,9	7,5	5,6	27,3	7,2	5,8	22,6	7,1	5,5	19,2	6,4	4,5	18,6	6,4	4,4	20,1	6,3	4,2
June	33,8	7,5	5,4	32,2	7,6	5,6	28,1	7,1	5,5	23,9	7,1	4,3	22,9	7,2	4,4	25,3	7,1	4,4
July	36,2	7,8	5,9	34,7	7,9	5,4	31,2	7,1	5,2	26,8	7,2	4,4	25,7	7,1	4,6	28,4	7,1	4,3
August	35,6	7,2	6,2	34,5	7,4	4,8	30,8	6,2	4,6	26,9	6,3	4,3	26,2	6,4	4,3	27,8	6,2	4,1
September	32,2	6,3	5,9	30,1	5,9	4,9	26,2	5,4	4,8	23,1	5,1	4,1	23,3	5,2	4,2	23,6	5,0	3,9
October	26,1	4,7	5,2	24,4	4,3	4,7	19,8	4,5	4,7	19,1	3,6	4,4	19,8	3,8	4,3	18,9	3,6	4,4
November	19,2	3,9	5,5	18,2	3,2	4,7	13,2	3,6	4,6	14,0	2,5	5,1	15,3	2,6	4,9	13,4	2,6	4,5
December	14,1	3,3	5,6	13,5	2,6	4,6	8,6	3,1	4,6	10,3	2,1	5,3	12,1	2,1	5,1	9,6	2,2	4,7
Average	24,8	5,8	5,6	23,3	5,5	5,1	19,1	5,3	4,9	17,3	4,6	4,7	17,7	4,7	4,7	17,7	4,6	4,5

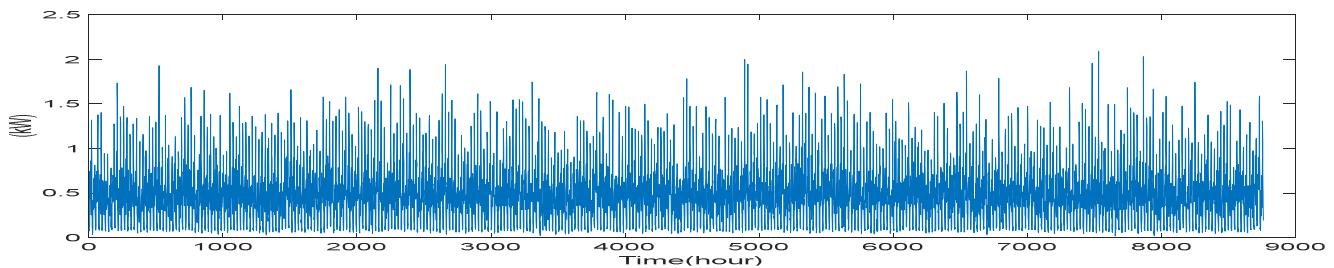


Fig. 3. Hourly household load profile over a year.

TABLE II. SPECIFICATIONS AND COSTS OF COMPONENTS USED IN THIS STUDY: PV PANEL, WIND TURBINE, BATTERY, AND CONVERTER.

Components	Details
PV panel	Maximum power 300 W
	Price 100 €

Battery lead-acid	Maintenance cost	2 €
	Lifetime (years)	20
	Price	700 €
	Nominal capacity/ voltage	260Ah/12V
	Efficiency	80%
	Lifetime	5 years

Wind turbine	Nominal power	650 W
	Price	2500 €
	Lifetime	20 years
	Cut in/out, rated speed	2.2/12/10m/s
converter	Price	2000 €
	Efficiency	96%
	Lifetime	10 years

The Table III provides a detailed comparison of configurations for hybrid energy systems across different provinces using the HOMER Pro software. Each province displays three primary configurations: PV/WT/Batt, PV/Batt, and WT/Batt, with corresponding values for the number of photovoltaic panels (N_{PV}), wind turbines (N_{WT}), and batteries (N_{Batt}). In the first configuration (PV/WT/Batt), we observe a higher number of solar panels for cities like Tiaret and Sétif. This can be explained by the lesser availability of solar energy in these regions compared to the desert areas, which are rich in solar power. Conversely, cities like Adrar and Algiers have fewer solar

panels, likely because of the utilization of two wind turbines, which reduce the dependency on solar power. For the second configuration (PV/Batt), there is a clear difference in the number of solar panels between northern and desert cities. Due to the higher solar irradiance in the southern desert cities, there is a decreased use of solar panels. In contrast, the opposite is observed for northern cities, which require more panels to compensate for the lower solar input. In the final configuration (WT/Batt), there's a noticeable similarity in the number of wind turbines in all provinces. This similarity can be linked to the comparable wind energy data we've seen across these regions. However, it's noteworthy that Adrar has the fewest wind turbines, indicating that the region is rich in wind energy potential. In essence, this table accentuates the geographical differences influencing energy choices. It's crucial for energy solutions to be region-specific, leveraging local resources optimally for sustainability and efficiency.

TABLE III. COMPARISON OF PV, WT, AND BATTERY CONFIGURATIONS IN VARIOUS PROVINCES.

province	PV/WT/Batt			PV/Batt		WT/Batt	
	N_{PV}	N_{WT}	N_{Batt}	N_{PV}	N_{Batt}	N_{WT}	N_{Batt}
Adrar	12	2	7	26	9	8	18
Ouargla	31	1	9	28	13	11	18
Béchar	23	1	7	26	9	9	24
Sétif	37	1	10	42	13	10	27
Algiers	27	2	8	36	13	11	25
Tiaret	40	1	8	38	11	13	24

TABLE IV. COMPARISON OF COE AND UNMET LOAD FOR DIFFERENT ENERGY CONFIGURATIONS IN VARIOUS PROVINCES.

province	PV/WT/Batt		PV/Batt		WT/Batt	
	COE (€/kWh)	Unmet load (%)	COE (€/kWh)	Unmet load (%)	COE (€/kWh)	Unmet load (%)
Adrar	0.503	0.009	0.522	0.025	1.03	0.046
Ouargla	0.557	0.015	0.572	0.024	1.2	0.062
Béchar	0.512	0.016	0.520	0.025	1.22	0.005
Sétif	0.582	0.01	0.603	0.022	1.33	0.013
Algiers	0.57	0.009	0.583	0.025	1.36	0.055
Tiaret	0.564	0.012	0.569	0.032	1.43	0.017

The Table IV provides a comparative analysis of the COE in euros per kilowatt-hour (€/kWh) and the percentage of unmet load for three different hybrid energy configurations with multiple provinces. From the data, it can be observed that the province of Adrar has the lowest COE for the PV/WT/Batt configuration at €0.503/kWh, with an unmet load percentage of 0.009%. In contrast, the province of Tiaret exhibits the highest COE in this configuration at €0.564/kWh but maintains a relatively low unmet load percentage at 0.012%. In the PV/Batt configuration, the COE ranges from €0.520/kWh in Béchar to €0.603/kWh in Sétif. The unmet load percentages in this configuration are slightly higher than the PV/WT/Batt configuration, with Tiaret having the highest percentage value of 0.032%. Lastly, the WT/Batt configuration shows a noticeable

increase in the COE values in all provinces. The province of Adrar has the lowest COE at €1.03/kWh, while Tiaret again records the highest at €1.43/kWh. The unmet load percentages in this configuration vary significantly, with Béchar having the lowest at 0.005% and Ouargla having a value of 0.062%. From these results, we can conclude that relying on a single energy source increases the system's cost. Consequently, the hybrid system aids in enhancing the system's reliability and reducing the cost. Additionally, meteorological data indicates that Algeria is suitable for the utilization of renewable energies due to the diversity and abundance of its energy sources. On the other hand, HOMER Pro did a great job sizing systems for different places, even with different weather data, making it a trusted

tool for sizing HESs. It also shows that we can make it even better by adding weather forecasts or using new algorithms.

IV. CONCLUSION

The paper introduces a simulated standalone hybrid power system model that integrates PV and Wind with a converter and battery storage. The system is evaluated in HOMER pro to obtain the best size of the HES. The research paper offers a comprehensive assessment of the viability of HES configurations in different provinces of Algeria. A variety of studies, evaluations, and conclusions have highlighted Algeria's potential for using renewable energy resources, particularly solar and wind. All three configurations provide promising results in different Algerian cities, with just some variance in terms of LCOE. It should be mentioned that, in this study, the PV/WT/Batt configuration produced the best results when compared to the other configuration. The highest value for unmet load is 0.046 percent, which means that all configurations can effectively and reliably meet the load demand over the long term. On the other hand, the results are also consistent with previous studies on the importance of renewable energy. This compels us to shift away from non-renewable energy sources in order to deal with the challenges posed by climate change and diminishing fossil fuel reserves. In this regard, the meteorological characteristics that characterize Algeria, which are characterized by their diversity and abundance, make it a suitable location for the use of renewable energy systems on a large scale, as seen in this study.

V. REFERENCES

- [1] H. Nasiraghdam and S. Jadid, "Optimal hybrid PV/WT/FC sizing and distribution system reconfiguration using multi-objective artificial bee colony (MOABC) algorithm," *Solar Energy*, vol. 86, no. 10, pp. 3057–3071, Oct. 2012, doi: 10.1016/j.solener.2012.07.014.
- [2] X. Xu, W. Hu, D. Cao, Q. Huang, C. Chen, and Z. Chen, "Optimized sizing of a standalone PV-wind-hydropower station with pumped-storage installation hybrid energy system," *Renewable Energy*, vol. 147, pp. 1418–1431, Mar. 2020, doi: 10.1016/j.renene.2019.09.099.
- [3] Z. Medghalchi and O. Taylan, "A novel hybrid optimization framework for sizing renewable energy systems integrated with energy storage systems with solar photovoltaics, wind, battery and electrolyzer-fuel cell," *Energy Conversion and Management*, vol. 294, p. 117594, Oct. 2023, doi: 10.1016/j.enconman.2023.117594.
- [4] N. Healy, J. C. Stephens, and S. A. Malin, "Embodied energy injustices: Unveiling and politicizing the transboundary harms of fossil fuel extractivism and fossil fuel supply chains," *Energy Research & Social Science*, vol. 48, pp. 219–234, Feb. 2019, doi: 10.1016/j.erss.2018.09.016.
- [5] M. Benmedjahed, A. Dahbi, A. Hadidi, and S. Mouhadjer, "Temperature and Wind Distribution Effects on Wind Energy Production in Adrar Region (Southern Algeria)," *IJSDP*, vol. 16, no. 8, pp. 1473–1477, Dec. 2021, doi: 10.18280/ijdp.160808.
- [6] F. Fodhil, A. Hamidat, and O. Nadjemi, "Potential, optimization and sensitivity analysis of photovoltaic-diesel-battery hybrid energy system for rural electrification in Algeria," *Energy*, vol. 169, pp. 613–624, Feb. 2019, doi: 10.1016/j.energy.2018.12.049.
- [7] O. Nadjemi, T. Nacer, A. Hamidat, and H. Salhi, "Optimal hybrid PV/wind energy system sizing: Application of cuckoo search algorithm for Algerian dairy farms," *Renewable and Sustainable Energy Reviews*, vol. 70, pp. 1352–1365, Apr. 2017, doi: 10.1016/j.rser.2016.12.038.
- [8] M. A. Mohamed, A. M. Eltamaly, A. I. Alolah, and A. Y. Hatata, "A novel framework-based cuckoo search algorithm for sizing and optimization of grid-independent hybrid renewable energy systems," *International Journal of Green Energy*, vol. 16, no. 1, pp. 86–100, Jan. 2019, doi: 10.1080/15435075.2018.1533837.
- [9] W. Zhang, A. Maleki, M. A. Rosen, and J. Liu, "Optimization with a simulated annealing algorithm of a hybrid system for renewable energy including battery and hydrogen storage," *Energy*, vol. 163, pp. 191–207, Nov. 2018, doi: 10.1016/j.energy.2018.08.112.
- [10] A. Tabak, E. Kayabasi, M. T. Guner, and M. Ozkaymak, "Grey wolf optimization for optimum sizing and controlling of a PV/WT/BM hybrid energy system considering TNPC, LPSP, and LCOE concepts," *Energy Sources, Part A: Recovery, Utilization, and Environmental Effects*, vol. 44, no. 1, pp. 1508–1528, Mar. 2022, doi: 10.1080/15567036.2019.1668880.
- [11] J. Lian, "A review on recent sizing methodologies of hybrid renewable energy systems," *Energy Conversion and Management*, 2019.
- [12] A. Kaabeche, M. Belhamel, and R. Ibtouen, "Sizing optimization of grid-independent hybrid photovoltaic/wind power generation system," *Energy*, vol. 36, no. 2, pp. 1214–1222, Feb. 2011, doi: 10.1016/j.energy.2010.11.024.
- [13] G. Zhang, C. Xiao, and N. Razmjoo, "Optimal operational strategy of hybrid PV/wind renewable energy system using homer: a case study," *International Journal of Ambient Energy*, vol. 43, no. 1, pp. 3953–3966, Dec. 2022, doi: 10.1080/01430750.2020.1861087.
- [14] Y. Basheer, A. Waqar, S. M. Qaisar, T. Ahmed, N. Ullah, and S. Alotaibi, "Analyzing the Prospect of Hybrid Energy in the Cement Industry of Pakistan, Using HOMER Pro," *Sustainability*, vol. 14, no. 19, p. 12440, Sep. 2022, doi: 10.3390/su141912440.
- [15] Sk. A. Shezan *et al.*, "Performance analysis of an off-grid wind-PV (photovoltaic)-diesel-battery hybrid energy system feasible for remote areas," *Journal of Cleaner Production*, vol. 125, pp. 121–132, Jul. 2016, doi: 10.1016/j.jclepro.2016.03.014.
- [16] S. W. Chisale, S. Eliya, and J. Taulo, "Optimization and design of hybrid power system using HOMER pro and integrated CRITIC-PROMETHEE II approaches," *Green Technologies and Sustainability*, vol. 1, no. 1, p. 100005, Jan. 2023, doi: 10.1016/j.grets.2022.100005.
- [17] K. Ram, P. K. Swain, R. Vallabhaneni, and A. Kumar, "Critical assessment on application of software for designing hybrid energy systems," *Materials Today: Proceedings*, vol. 49, pp. 425–432, 2022, doi: 10.1016/j.matpr.2021.02.452.
- [18] Md. Islam, H. Akter, H. Howlader, and T. Senju, "Optimal Sizing and Techno-Economic Analysis of Grid-Independent Hybrid Energy System for Sustained Rural Electrification in Developing Countries: A Case Study in Bangladesh," *Energies*, vol. 15, no. 17, p. 6381, Sep. 2022, doi: 10.3390/en15176381.
- [19] M. A. Zaman and Md. A. Razzak, "Sustainable Microgrid Analysis for Kutubdia Island of Bangladesh," *IEEE Access*, vol. 10, pp. 37533–37556, 2022, doi: 10.1109/ACCESS.2022.3164677.
- [20] V. Suresh, M. M., and R. Kiranmayi, "Modelling and optimization of an off-grid hybrid renewable energy system for electrification in a rural areas," *Energy Reports*, vol. 6, pp. 594–604, Nov. 2020, doi: 10.1016/j.eegy.2020.01.013.